

**Privatization's Unequal Toll: Explaining Cross-Country
Variation in Mortality and Health Outcomes After Transition
From Communism**

Introduction

The collapse of communism in Central and Eastern Europe as well as in the former Soviet Union in the late 1980s and early 1990s marked a significant turning point in global political and economic history. The complexity of carrying out political, territorial, and economic reforms simultaneously is often termed the "dilemma of synchronicity" [1]. This concept underscores the immense challenge faced by these nations in attempting to democratize political systems, redefine state borders or identities, and restructure economies concurrently, rather than sequentially. A cornerstone of this economic paradigm shift was privatization, which entailed the replacement of planned economies with free-market principles [2]. This process targeted a diverse range of state-owned assets, including both enterprises established as public property and those that had been previously private before being nationalized [1]. Furthermore, the execution of privatization varied significantly across the region; while some countries adopted rapid "shock therapy" or mass voucher schemes, others pursued a more gradualist approach favoring direct sales or restitution [3, 4]. That process was expected to lead to increased efficiency, economic growth, and improved living standards [5]. However, the outcomes of this transition have been far from uniform across countries, with some nations experiencing significant improvements in health and mortality rates, while others have faced deteriorating conditions [6].

This paper aims at ...

Relevant Literature

Transition Types

The widespread institutional collapse across Central and Eastern Europe and the Former Soviet Union in the late 1980s led to an imperative for simultaneous political, territorial, and economic restructuring. Within the academic field of transitology, literature seeks to define

and categorize these complex processes in various ways [7]. Probably the most commonly referenced typology of transformation processes [7] of Samuel Huntington divides countries into three categories based on the nature of their reforms [8]:

- "transformation" (e.g., Soviet Union, Bulgaria, Hungary) - when reformators group is originated in government elites,
- "transplacement" (e.g., Czechoslovakia, Poland) - when the reforms are imposed by both government elites and opposition forces, as it happened during the round table talks,
- "replacement" (e.g., East Germany, Romania) - when opposition has a significant role in reforms.

These categories reflect the degree of domestic consensus and external pressure influencing the reform processes. It is also worth mentioning that both power camp and opposition are divided, government into reformators and "hard-headed", opposition into moderates and radicals. In case of other classifications, they mostly match the above-mentioned typology, sometimes however some differences in countries grouping can be observed. Sometimes, a separate category is being created for countries created from multi-national states, like Czechoslovakia or Yugoslavia, due to merged effect of divisions on the level of top leadership and the nationalistic pressure from below [9]. Worth to mention is also the division proposed by Herbert Kitschelt due to underlining group of countries where preemptive reforms were the idea. Examples are Soviet Union Gorbachev's initial innovations, as well as the regime changes in Bulgaria, Romania etc. [10]. The variation in the transition yields the opportunity to examine its elements' impact on the outcomes, specifically mortality.

Economic Transformation and the Privatization Debate

Privatization was universally deemed a central component of the economic transformation, necessary to replace the inefficient command economy with market principles [11, 12].

The public debate regarding the pace and method of implementation was dominated by a dichotomy between two camps. The first camp consisted of neoliberal advisors who championed "shock therapy" advocating for the rapid and simultaneous implementation of price and trade liberalization, stabilization, and mass privatization [13]. This approach was intended to create an irreversible shift to the market. In opposition to them stood critics, often described as gradualist economists or neo-institutionalists, who argued that hasty reform damaged the state and favored a slower process to allow sufficient time for essential governing institutions (e.g. corporate law, capital markets) to develop [4, 14].

The specific privatization methods chosen across Central and Eastern Europe and the Former Soviet Union yielded varied results and attracted distinct academic critiques. Regarding mass (voucher) privatization, while econometric results suggest that it generally had a significant positive effect on growth following the initial recession (after 1995) [2], its implementation faced scrutiny. Particularly in the Czech Republic, this method was criticized for leading to "budgetary crises and unleashing inflation at the macroeconomic level" [15]. Conversely, insider privatization models that favored management and employees, such as those in the Russian Federation and Ukraine, resulted in "insider dominance" and were generally less effective at restructuring than market methods [2, 16]. This approach was associated with political continuity and was mostly restricted in the Czech Republic and Hungary, and to a lesser extent in Poland, where a hard political transition had weakened the positions of former communist managers [3].

Beyond specific methodologies, the literature confirms that the transition created new incentives for corruption due to the massive transfer of property coupled with weak states and underdeveloped civil societies [17, 18, 19]. Research indicates that privatization's general growth effect hinges on the quality of the institutional setting, often proxied by the level of corruption [20]. Furthermore, the implementation of mass privatization programs has been empirically demonstrated to create a massive fiscal shock for post-communist governments, which undermined state capacity and the protection of property rights, thereby exacerbating

the recession [13]. Ultimately, the general dissatisfaction with the privatization process stemmed from the realization of the process and its outcomes rather than opposition to private property itself [21]. Consequently, the popular demand for revising privatization was strongly correlated with individuals who experienced hardships during the transition, such as wage cuts, unemployment, or poor self-rated health.

Building on this, the literature suggests that the institutional setting acts as a critical moderator of privatization’s toll [20, 22]. While mass privatization can undermine state capacity [13], the presence of robust governance and active civil society participation may serve as a buffer against the most disruptive social effects [14, 23]. This perspective shifts the focus from the mere pace of reform to the structural environment in which property transfer occurs [24], suggesting that countries with higher levels of civic engagement and better enterprise restructuring oversight might mitigate the transition’s negative health consequences [14, 25].

Mortality and Health Outcomes

The political and economic upheaval led to a severe post-socialist mortality crisis between 1989 and 1995, resulting in an estimated 10 million excess deaths between 1990 and 2000 [23]. However, different countries experienced divergent trajectories in health outcomes:

- **Severe Crisis:** FSU states, including Russian Federation, Ukraine, and Belarus, experienced dramatic and prolonged mortality reversals [26]. Russian Federation male life expectancy, for example, dropped by 6.6 years between 1989 and 1994, reaching a level three years below that of India, which was then a poor agrarian economy [5]. The death toll in post-Soviet states reached an estimated 7 million premature deaths, with 4 million in Russian Federation alone [27].
- **Rapid Recovery:** Central European countries, such as the Czech Republic and Poland, were the first former Communist countries to see a “rapid and sustained increase in life expectancy” after 1991 [28].

To understand the drivers of these demographic shifts, it is necessary to examine the specific structure of mortality. The literature identifies the crisis as being primarily fueled by a surge in cardiovascular diseases (particularly ischemic heart disease) and external causes (accidents, suicide, and alcohol poisoning) [23, 27].

This excess mortality was concentrated among working-age men, though with distinct patterns across cohorts. The surge was especially pronounced among young adults (20–39), who were predominantly affected by external causes, and older working-age adults (40–59), who suffered mainly from a rise in cardiovascular deaths [5]. In the Russian Federation, for instance, the fastest relative upswing occurred in the younger group, while the largest absolute rise was recorded among the older cohort [26].

These specific medical outcomes bridge the gap between economic policy and demographic data. They serve as biological markers of acute psychosocial stress triggered by rapid economic dislocation and unemployment [14]. This stress manifests precisely through the channels observed above: physiological strain leading to heart failure [6, 29] and psychological distress resulting in “deaths of despair” [30]. By focusing on these indicators, the analysis isolates the direct physiological and social toll of privatization policies.

Despite the moderating role of institutions, a fundamental division persists in the literature regarding the extent to which mass privatization itself was the decisive upstream factor driving mortality crises. The arguments can be grouped as follows:

- **Argument for Correlation (Privatization as a Crucial Factor):** A large body of cross-national and quasi-experimental research supports the hypothesis that rapid and extensive privatization (often referred to as mass privatization) was a “crucial determinant” of the adverse mortality trends [14, 27, 30, 31]. The primary mechanism linking this policy to premature death is identified as acute psychosocial stress caused by rapid economic dislocation. For example, one quasi-experimental retrospective cohort study in post-Soviet Russian Federation found that fast-privatized towns experienced 13% higher mortality among working-age men than slow-privatized towns [27]. These

findings provided compelling evidence that rapid privatization contributed to raised working-age male mortality. In case of female related research, the one was made among women in Hungary [32]. Results indicated that prolonged state ownership was associated with protection of life chances during the post-socialist transformation for women. Another research reexamined the argument that alcohol policies were the key behind the mortality crisis in Russian Federation, and they found it as non robust. At the same time, they confirmed the robust link between rapid privatization and increased mortality [31]. This link is further supported by evidence that privatization, and the associated increase in unemployment, was strongly associated with mortality in former Soviet Union countries [14].

- **Argument Against Robust Correlation (Methodological Skepticism):** Conversely, other prominent studies challenge the conclusion that mass privatization was the primary cause of the crisis, citing methodological weaknesses and a lack of empirical robustness [33, 34, 35, 36]. Critics argue that the claim that mass privatization adversely affected male mortality trends "does not stand up to closer examination" and is contradicted by other knowledge about health trends in the region. Some analyses found that the correlation between rapid privatization and mortality fails to hold across countries. Specifically, a study focusing on Russian Federationn regions found no robust evidence supporting the hypotheses that the mortality spike was caused primarily by the rebound effect (from the Gorbachev anti-alcohol campaign) or the affordability of alcohol. Instead, they concluded that the privatization-mortality link is robust to controlling for alcohol policy [37]. Other reviews, however, found that while four out of five studies showed rapid privatization to be significantly and positively related to mortality [38], with the only exception of Christopher J. Gerry [39]. Gerry critiques the Stuckler et al. article for committing an ecological fallacy. He argues that their broad, country-level analysis fails to prove that mass privatization was directly responsible for causing death at the individual level.

The overall scientific knowledge on the effects of politico-economic transitions on health outcomes remains "inconclusive" and there definitely is an area to make further examination, which the rest of the paper will be devoted to.

Data and Methods

Data Sources and Study Population

To examine the health consequences of structural transformation, we constructed a longitudinal cross-country dataset covering 14 post-communist economies in Central and Eastern Europe (CEE) and the Former Soviet Union (FSU) over the period 1989–2014. The final sample constitutes a balanced panel including Belarus, Bulgaria, Czech Republic, Estonia, Hungary, Kazakhstan, Latvia, Lithuania, Poland, Romania, Russian Federation, Slovakia, Slovenia, and Ukraine.

The dataset integrates demographic and mortality data from the **World Health Organization (WHO) European Health for All Database (HFA-DB)** [40]. To capture the economic and institutional dimensions of transition, we draw on structural reform indicators from the **European Bank for Reconstruction and Development (EBRD)** [41] and historically reconstructed GDP series from the **Maddison Project Database** [42].

Complementary data on social conditions, institutional quality, and state capacity were obtained from the **Varieties of Democracy (V-Dem)** project [43], the **World Bank's World Development Indicators (WDI)** [44], and **OECD Health Data** [45]. Additional labor market and social expenditure statistics were sourced from **UNICEF** databases and reports [46, 47]. Supplemental macroeconomic controls (inflation, private sector share) utilize historical data from EBRD Transition Reports [25, 48, 49].

The study population represents distinct transition trajectories, geographically and institutionally clustered into Central Europe (Poland, Czech Republic, Slovakia, Hungary), the Baltic states (Estonia, Latvia, Lithuania), South-Eastern Europe (Bulgaria, Romania,

Slovenia), and the FSU (Russian Federation, Ukraine, Belarus, Kazakhstan). The temporal scope captures the immediate pre-transition baseline (1989), the turbulent early transition years characterized by transformational recession, and the subsequent stabilization phases up to 2014.

Measures

Outcome Variables: Mortality and Health

Population health is operationalized using Standardized Death Rates (SDR) per 100,000 population. To disentangle the specific mechanisms linking economic shock to mortality, a broad spectrum of cause-specific outcomes available in the WHO HFA-DB is analyzed.

First, to capture the physiological toll of acute psychosocial stress, the **SDR from Ischaemic Heart Disease** is employed. Cardiovascular mortality is widely recognized as highly responsive to rapid social disruptions [23, 29]. Second, the **SDR from External Causes of Injury and Poisoning** is examined. This aggregate category serves as a proxy for societal instability and violence [14]. Within this domain, the **SDR from Suicide and Self-inflicted Injury** is specifically isolated to directly test the impact of transition on psychological distress and social anomie [50, 30].

Furthermore, given the prominence of the “alcohol hypothesis” in the literature, the **SDR from Alcohol-related Causes** is included to verify whether mortality spikes were driven primarily by hazardous drinking behaviors [37, 51, 36]. Finally, consistent with rigorous methodological standards, the **SDR from Malignant Neoplasms** is employed as a negative control (placebo outcome). Since cancer mortality is driven by long-term etiological processes, it should theoretically remain unresponsive to short-term economic shocks [52]. **Life Expectancy at Birth** is also reported as an aggregate measure of population welfare.

Explanatory Variables: The Multidimensional Transition Shock

To quantify the intensity of structural reforms, the study utilizes the complete set of transition indicators provided by the EBRD. While the primary focus lies on ownership transformation, the multidimensional nature of the market shock is accounted for by including indices for:

- **Large-scale Privatization:** Progress in the transfer of large state assets to private hands.
- **Small-scale Privatization:** Progress in the privatization of small-scale assets.
- **Governance and Enterprise Restructuring:** Progress in hardening budget constraints and corporate governance.
- **Price Liberalization:** Progress in lifting price controls.
- **Trade and Forex System:** Liberalization of foreign trade and currency exchange.
- **Competition Policy:** Development of anti-monopoly legislation and institutions.

All EBRD indices are measured on a scale from 1.0 (centrally planned economy) to 4.3 (advanced market economy) [53]. However, literature suggests that these subjective indices may be prone to perception bias and fail to distinguish between different methods of property transfer [54]. Therefore, complementing these expert assessments, the **Private Sector Share of GDP** is included as an objective, outcome-based measure. Furthermore, a dummy variable for **Mass Privatization** is constructed to explicitly test the impact of voucher-based schemes, which are theoretically distinct from gradual sales methods.

Covariates and Moderators

The model accounts for a rich set of time-varying confounders. To isolate the effect of reforms from economic development, the natural logarithm of **Real GDP per capita** (in 2011 constant

US\$, from the Maddison Project) is controlled for. As a proxy for labor market security and economic inclusion—crucial during the restructuring period—the **Employment Rate** (sourced from UNICEF TransMONEE) is employed, offering superior data completeness compared to unemployment statistics for the early 1990s. Macroeconomic instability is also controlled for using the annual **Inflation Rate**.

To test the buffering role of institutions, the **Civil Society Participation Index** (from V-Dem) is incorporated to proxy for social capital, alongside **Health Equality** to measure the fairness of health protection. State capacity is captured by **Government Consumption Expenditure** (% of GDP). Finally, to account for behavioral risks and exogenous shocks, average **Alcohol Consumption** (liters per capita) is included, and a binary variable for **War** is constructed to adjust for armed conflicts.

Table 1 provides a summary of all variables included in the dataset.

Table 1: List of Variables and Data Sources

Variable Category		Specific Indicators Included
Health Outcomes (WHO)		Life Expectancy, SDR Ischaemic Heart Disease, SDR External Causes, SDR Suicide, SDR Alcohol-related Causes, SDR Malignant Neoplasms (Placebo).
Transition (EBRD)	Indicators	Large-scale Privatization, Small-scale Privatization, Governance & Restructuring, Price Liberalization, Trade & Forex System, Competition Policy.
Economic Structure		Private Sector Share of GDP (EBRD), Employment Rate (UNICEF), Inflation Rate (EBRD).
Macro-Controls		Real GDP per Capita (Maddison Project), Gov. Consumption Expenditure (WB WDI).
Social & Institutional		Civil Society Participation (V-Dem), Health Equality (V-Dem), Alcohol Consumption (OECD), War (Dummy).

Results

Conclusion

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